SE : LAB 7: Fuzzing for Input Validation Bugs

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SECTION: 5 J

SS1: buggy script crashing on abc and 1200/0

Run 1: ValueError: invalid literal for int() with base 10: 'abc'

Run 2: ZeroDivisionError: integer division or modulo by zero

```
PS C:\Users\DARSHANA\Pictures\SEM 5\SE\lab 6\L6_L7_58_restaurant_bill\LAB7> python restaurant_bill_parser.py
==== Running clean_dish_name ====
Enter a dish name from the menu (e.g. *paneer tikka*): *paneer tikka*
Dish name: Paneer Tikka

==== Running parse_item_prices ====
Enter item prices separated by commas (e.g. 250,180,320): 250,abc
Traceback (most recent call last):
  File "C:\Users\DARSHANA\Pictures\SEM 5\SE\lab 6\L6_L7_58_restaurant_bill\LAB7\restaurant_bill_parser.py", line 40, in <module>
    main()
  File "C:\Users\DARSHANA\Pictures\SEM 5\SE\lab 6\L6_L7_58_restaurant_bill\LAB7\restaurant_bill_parser.py", line 33, in main
    print("Item prices:", parse_item_prices(value))
                          ^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\DARSHANA\Pictures\SEM 5\SE\lab 6\L6_L7_58_restaurant_bill\LAB7\restaurant_bill_parser.py", line 16, in parse_item_prices
    return [int(price) for price in text.split(',')]
            ^^^^^^^^^^
ValueError: invalid literal for int() with base 10: 'abc'
PS C:\Users\DARSHANA\Pictures\SEM 5\SE\lab 6\L6_L7_58_restaurant_bill\LAB7>
PS C:\Users\DARSHANA\Pictures\SEM 5\SE\lab 6\L6_L7_58_restaurant_bill\LAB7>
PS C:\Users\DARSHANA\Pictures\SEM 5\SE\lab 6\L6_L7_58_restaurant_bill\LAB7> python restaurant_bill_parser.py
==== Running clean_dish_name ====
Enter a dish name from the menu (e.g. *paneer tikka*): *paneer tikka*
Dish name: Paneer Tikka

==== Running parse_item_prices ====
Enter item prices separated by commas (e.g. 250,180,320): 250,180,320
Item prices: [250, 180, 320]

==== Running split_amount ====
Enter the bill and number of people as TOTAL/PEOPLE (e.g. 1200/4): 1200/0
Traceback (most recent call last):
  File "C:\Users\DARSHANA\Pictures\SEM 5\SE\lab 6\L6_L7_58_restaurant_bill\LAB7\restaurant_bill_parser.py", line 40, in <module>
    main()
  File "C:\Users\DARSHANA\Pictures\SEM 5\SE\lab 6\L6_L7_58_restaurant_bill\LAB7\restaurant_bill_parser.py", line 37, in main
    print("Each person pays:", split_amount(value))
                               ^^^^^^^^^^^^^^^^^^^
  File "C:\Users\DARSHANA\Pictures\SEM 5\SE\lab 6\L6_L7_58_restaurant_bill\LAB7\restaurant_bill_parser.py", line 24, in split_amount
    return int(total) // int(people)
           ~~~~~~~~~~~^^~~~~~~~~~~~~
ZeroDivisionError: integer division or modulo by zero
PS C:\Users\DARSHANA\Pictures\SEM 5\SE\lab 6\L6_L7_58_restaurant_bill\LAB7>
```

SS2: the Hypothesis tests failing on the buggy code

```
PS C:\Users\DARSHANA\Pictures\SEM 5\SE\lab 6\L6_L7_58_restaurant_bill\LAB7> python -m pytest test_restaurant_bill_parser.py
============================= test session starts ==============================
platform win32 -- Python 3.12.10, pytest-9.2.0, pluggy-1.7.0
rootdir: C:\Users\DARSHANA\Pictures\SEM 5\SE\lab 6\L6_L7_58_restaurant_bill\LAB7
plugins: anyio-4.10.0, hypothesis-6.170.15
collected 3 items

test_restaurant_bill_parser.py FFF                                       [100%]

=================================== FAILURES ===================================
_____________________________ test_clean_dish_name_no_crash _____________________________

s = None

    @given(st.text() | st.none())
    def test_clean_dish_name_no_crash(s):
        try:
>           clean_dish_name(s)

test_restaurant_bill_parser.py:8:
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

name = None

    def clean_dish_name(name):
        """
        Removes the chef's-special markers (*, #, ~) and title-cases the name.
        Assumes name is a non-empty string.
        """
>       name = name.strip()
               ^^^^^^^^^
E       AttributeError: 'NoneType' object has no attribute 'strip'

restaurant_bill_parser.py:6: AttributeError

During handling of the above exception, another exception occurred:

    @given(st.text() | st.none())
>   def test_clean_dish_name_no_crash(s):
                    ^^^

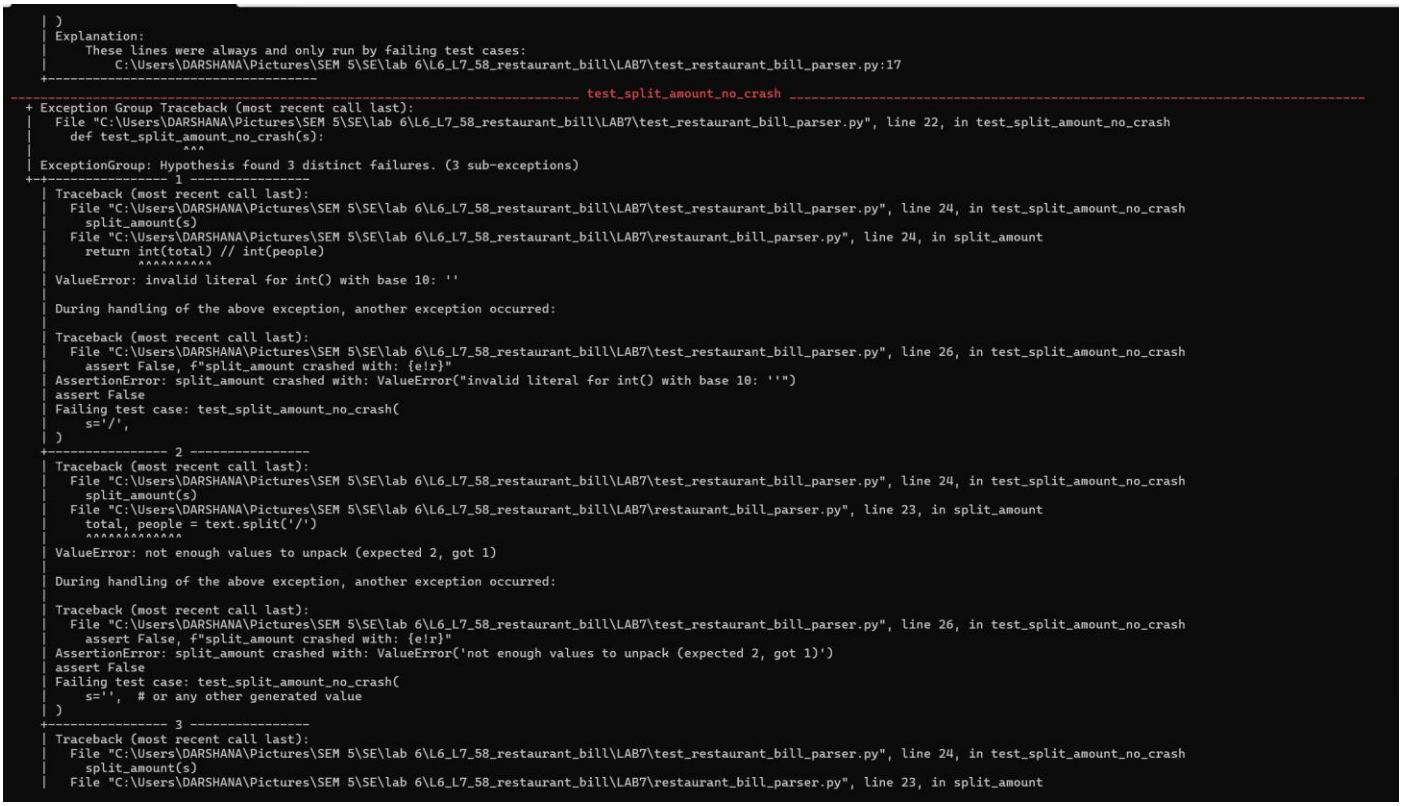
test_restaurant_bill_parser.py:6:
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

s = None

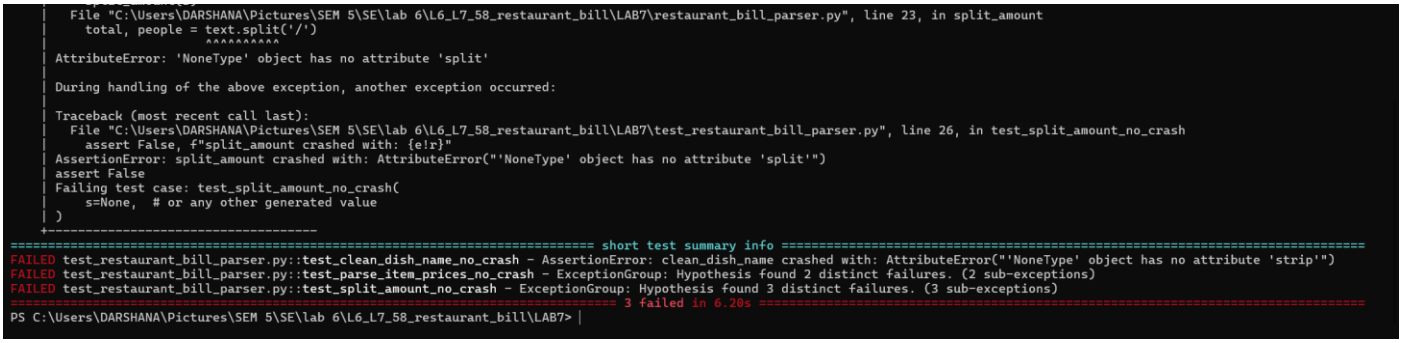
    @given(st.text() | st.none())
    def test_clean_dish_name_no_crash(s):
        try:
            clean_dish_name(s)
        except Exception as e:
>           assert False, f"clean_dish_name crashed with: {e!r}"
E           AssertionError: clean_dish_name crashed with: AttributeError("'NoneType' object has no attribute 'strip'")
E           assert False
```

```
E           Failing test case: test_clean_dish_name_no_crash(
E               s=None,
E           )
E           Explanation:
E               These lines were always and only run by failing test cases:
E                   C:\Users\DARSHANA\Pictures\SEM 5\SE\lab 6\L6_L7_58_restaurant_bill\LAB7\test_restaurant_bill_parser.py:9

test_restaurant_bill_parser.py:10: AssertionError
_____________________________ test_parse_item_prices_no_crash _____________________________
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\DARSHANA\Pictures\SEM 5\SE\lab 6\L6_L7_58_restaurant_bill\LAB7\test_restaurant_bill_parser.py", line 14, in test_parse_item_prices_no_crash
  |     def test_parse_item_prices_no_crash(s):
  |                    ^^^
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\DARSHANA\Pictures\SEM 5\SE\lab 6\L6_L7_58_restaurant_bill\LAB7\test_restaurant_bill_parser.py", line 16, in test_parse_item_prices_no_crash
    |     parse_item_prices(s)
    |   File "C:\Users\DARSHANA\Pictures\SEM 5\SE\lab 6\L6_L7_58_restaurant_bill\LAB7\restaurant_bill_parser.py", line 16, in parse_item_prices
    |     return [int(price) for price in text.split(',')]
    |                ^^^^^^^^^
    | ValueError: invalid literal for int() with base 10: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\DARSHANA\Pictures\SEM 5\SE\lab 6\L6_L7_58_restaurant_bill\LAB7\test_restaurant_bill_parser.py", line 18, in test_parse_item_prices_no_crash
    |     assert False, f"parse_item_prices crashed with: {e!r}"
    | AssertionError: parse_item_prices crashed with: ValueError("invalid literal for int() with base 10: ''")
    | assert False
    | Failing test case: test_parse_item_prices_no_crash(
    |     s='',
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         C:\Users\DARSHANA\Pictures\SEM 5\SE\lab 6\L6_L7_58_restaurant_bill\LAB7\test_restaurant_bill_parser.py:17
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\DARSHANA\Pictures\SEM 5\SE\lab 6\L6_L7_58_restaurant_bill\LAB7\test_restaurant_bill_parser.py", line 16, in test_parse_item_prices_no_crash
    |     parse_item_prices(s)
    |   File "C:\Users\DARSHANA\Pictures\SEM 5\SE\lab 6\L6_L7_58_restaurant_bill\LAB7\restaurant_bill_parser.py", line 16, in parse_item_prices
    |     return [int(price) for price in text.split(',')]
    |                                     ^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\DARSHANA\Pictures\SEM 5\SE\lab 6\L6_L7_58_restaurant_bill\LAB7\test_restaurant_bill_parser.py", line 18, in test_parse_item_prices_no_crash
    |     assert False, f"parse_item_prices crashed with: {e!r}"
    | AssertionError: parse_item_prices crashed with: AttributeError("'NoneType' object has no attribute 'split'")
    | assert False
    | Failing test case: test_parse_item_prices_no_crash(
    |     s=None,
```



```
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         C:\Users\DARSHANA\Pictures\SEM 5\SE\lab 6\L6_L7_58_restaurant_bill\LAB7\test_restaurant_bill_parser.py:17
    +------------------------------------
_____ test_split_amount_no_crash _____
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\DARSHANA\Pictures\SEM 5\SE\lab 6\L6_L7_58_restaurant_bill\LAB7\test_restaurant_bill_parser.py", line 22, in test_split_amount_no_crash
  |     def test_split_amount_no_crash(s):
  |                    ^^^
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\DARSHANA\Pictures\SEM 5\SE\lab 6\L6_L7_58_restaurant_bill\LAB7\test_restaurant_bill_parser.py", line 24, in test_split_amount_no_crash
    |     split_amount(s)
    |   File "C:\Users\DARSHANA\Pictures\SEM 5\SE\lab 6\L6_L7_58_restaurant_bill\LAB7\restaurant_bill_parser.py", line 24, in split_amount
    |     return int(total) // int(people)
    |            ^^^^^^^^^^
    | ValueError: invalid literal for int() with base 10: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\DARSHANA\Pictures\SEM 5\SE\lab 6\L6_L7_58_restaurant_bill\LAB7\test_restaurant_bill_parser.py", line 26, in test_split_amount_no_crash
    |     assert False, f"split_amount crashed with: {e!r}"
    | AssertionError: split_amount crashed with: ValueError("invalid literal for int() with base 10: ''")
    | assert False
    | Failing test case: test_split_amount_no_crash(
    |     s='/',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\DARSHANA\Pictures\SEM 5\SE\lab 6\L6_L7_58_restaurant_bill\LAB7\test_restaurant_bill_parser.py", line 24, in test_split_amount_no_crash
    |     split_amount(s)
    |   File "C:\Users\DARSHANA\Pictures\SEM 5\SE\lab 6\L6_L7_58_restaurant_bill\LAB7\restaurant_bill_parser.py", line 23, in split_amount
    |     total, people = text.split('/')
    |     ^^^^^^^^^^^^^
    | ValueError: not enough values to unpack (expected 2, got 1)
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\DARSHANA\Pictures\SEM 5\SE\lab 6\L6_L7_58_restaurant_bill\LAB7\test_restaurant_bill_parser.py", line 26, in test_split_amount_no_crash
    |     assert False, f"split_amount crashed with: {e!r}"
    | AssertionError: split_amount crashed with: ValueError('not enough values to unpack (expected 2, got 1)')
    | assert False
    | Failing test case: test_split_amount_no_crash(
    |     s='',  # or any other generated value
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\DARSHANA\Pictures\SEM 5\SE\lab 6\L6_L7_58_restaurant_bill\LAB7\test_restaurant_bill_parser.py", line 24, in test_split_amount_no_crash
    |     split_amount(s)
    |   File "C:\Users\DARSHANA\Pictures\SEM 5\SE\lab 6\L6_L7_58_restaurant_bill\LAB7\restaurant_bill_parser.py", line 23, in split_amount
```



```
    |   File "C:\Users\DARSHANA\Pictures\SEM 5\SE\lab 6\L6_L7_58_restaurant_bill\LAB7\restaurant_bill_parser.py", line 23, in split_amount
    |     total, people = text.split('/')
    |     ^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\DARSHANA\Pictures\SEM 5\SE\lab 6\L6_L7_58_restaurant_bill\LAB7\test_restaurant_bill_parser.py", line 26, in test_split_amount_no_crash
    |     assert False, f"split_amount crashed with: {e!r}"
    | AssertionError: split_amount crashed with: AttributeError("'NoneType' object has no attribute 'split'")
    | assert False
    | Failing test case: test_split_amount_no_crash(
    |     s=None,  # or any other generated value
    | )
    +------------------------------------
=========================== short test summary info ============================
FAILED test_restaurant_bill_parser.py::test_clean_dish_name_no_crash - AssertionError: clean_dish_name crashed with: AttributeError("'NoneType' object has no attribute 'strip'")
FAILED test_restaurant_bill_parser.py::test_parse_item_prices_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_restaurant_bill_parser.py::test_split_amount_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
============================== 3 failed in 6.20s ===============================
PS C:\Users\DARSHANA\Pictures\SEM 5\SE\lab 6\L6_L7_58_restaurant_bill\LAB7>
```

SS3: fix the module

The fixed script handled all three bad inputs without crashing:

paneer tikka → Paneer Tikka

250,abc,,-5,320 → [250, 320]

1200/0 → a clear “Invalid input” message

```
PS C:\Users\DARSHANA\Pictures\SEM 5\SE\lab 6\L6_L7_58_restaurant_bill\LAB7> python restaurant_bill_parser.py
==== Running clean_dish_name ====
Enter a dish name from the menu (e.g. *paneer tikka*): *paneer tikka*
Dish name: Paneer Tikka

==== Running parse_item_prices ====
Enter item prices separated by commas (e.g. 250,180,320): 250,abc,,-5,320
Item prices: [250, 320]

==== Running split_amount ====
Enter the bill and number of people as TOTAL/PEOPLE (e.g. 1200/4): 1200/0
Invalid input: use TOTAL/PEOPLE with whole numbers and at least 1 person.
PS C:\Users\DARSHANA\Pictures\SEM 5\SE\lab 6\L6_L7_58_restaurant_bill\LAB7>
```

SS4: run the tests on the fixed module

